TUSHANG PATEL

Embedded Firmware Engineer | STM32 · ESP32 · FreeRTOS · LTE/BLE IoT

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SUMMARY

Embedded Firmware Engineer with 2+ years shipping production firmware for **neonatal medical devices, EV power systems, motor drives, and LTE/BLE IoT products**. Own the full firmware lifecycle: bring-up, RTOS design, drivers, fail-safe bootloaders/OTA, cloud telemetry, and field debugging.

TECHNICAL SKILLS

Languages & RTOS: Embedded C, C++, Python, MicroPython / QuecPython; FreeRTOS, CMSIS-RTOS, ESP-IDF; tasks, queues, mutexes, semaphores, event groups, ISR-safe design, DMA, low-power design (deep/light sleep, motion wake-up)

MCUs & SoCs: STM32 (H7, H5, G4, G0, F4), ESP32 (P4, C6, C3, WROVER), Puya PY32F002B, ATmega328P/2560, Quectel EC200U (LTE Cat-1) and MC60 (GSM/GNSS)

Protocols: BLE (NimBLE), LTE/GSM, GNSS/AGPS, Modbus RTU (RS-485), MQTT/MQTTS, HTTP, WebSockets, USB Host (TinyUSB), SPI, I2C, UART, I2S

Motor Control & Hardware: BLDC/PMSM FOC (ST MCSDK), 6-step, Hall sensors, stepper profiling, ADC/DAC, PWM timers, OPAMP/COMP, SPI Flash, FATFS, LiDAR, ADXL345, RC522 RFID

Bootloaders & Security: Custom dual-stage bootloader, SoftAP web OTA, partition tables, rollback protection, SHA-256 image verification, secure boot flow

Tools: STM32CubeIDE/MX/Programmer, ESP-IDF, Keil µVision, VS Code, Git/GitHub/GitLab, oscilloscope (DSO), J-Link / ST-Link

WORK EXPERIENCE

Embedded Firmware Engineer | OOB Services, Ahmedabad *Jan 2025 – Present*

- Own end-to-end firmware for multiple products across STM32, ESP32, PY32, ATmega and Quectel LTE modems, from schematic review and bring-up to field deployment.
- Built connected IoT products (EC200U LTE, MC60 GSM/GPS, ESP32-C3 BLE) with encrypted MQTT telemetry to Azure IoT Hub and AWS IoT, plus offline-first caching so no data is lost in weak coverage.
- Designed dual-stage bootloaders and SoftAP web OTA with isolated partitions, so field updates cannot brick devices.
- Wrote FreeRTOS firmware with queue-driven task communication and ISR-safe memory handling; validated signals with oscilloscope and worked with hardware engineers to fix integration issues before production.

Embedded Firmware Engineer | Electrogenics Innovations, Ahmedabad *Jul 2024 – Dec 2024*

- Developed STM32 firmware for regulated neonatal devices (Bubble CPAP, Vital Signs Monitor) aligned with IEC 60601-1-8 alarm requirements.
- Implemented FreeRTOS multitasking and drivers (UART, SPI, I2C, ADC+DMA) for deterministic patient-safety monitoring.
- Designed PPG signal-processing filters (bandpass, DC tracking, motion-artifact rejection) for SpO2 and heart-rate measurement.
- Created a reusable W25Q16BV SPI Flash driver library adopted across multiple internal projects.

KEY PROJECTS

3-Phase BLDC/PMSM FOC Motor Controller | *STM32G431 (Cortex-M4F, 170 MHz)*

- Built Field Oriented Control with ST MCSDK v6.4.2, Hall feedback and FreeRTOS, using hardware **CORDIC** for trig math and on-chip **OPAMPs** for phase-current sensing.
- Added a sub-50 ns hardware overcurrent trip by wiring internal comparators to TIM1 Break; added Modbus RTU slave and a Python GUI for live tuning.

Train Audio Announcer with Custom Bootloader & Web OTA | *ESP32-P4 / ESP32-C6 / WROVER*

- Built GPS geofence-triggered audio announcements (I2S, MP3 from SD/FATFS) with an LVGL v9.5 dashboard and a Wi-Fi portal for crew to upload audio and edit routes.
- Designed a fail-safe bootloader (factory / ota_0 / ota_1, key-gated recovery) with SoftAP browser OTA; a FreeRTOS queue worker buffers 512-byte chunks to avoid socket timeouts and stack overflows.
- Ported to **ESP32-P4 (dual-core RISC-V, 400 MHz)** with an ESP32-C6 Wi-Fi 6 coprocessor over SDIO.

Smart Bicycle Lock & Anti-Theft Sentinel | *EC200U + ESP32-C3 / MC60*

- Linked an ultra-low-power BLE bridge (ESP32-C3, deep-sleep with ADXL345 motion wake-up) with an LTE/GNSS engine (EC200U); built session-token BLE authentication and motor bolt control with jam detection.
- Cut GPS time-to-first-fix **91%** (90 s to under 8 s) on MC60 with an over-the-air AGPS (EPO) ephemeris pipeline.

High-Voltage EV Power Distribution & Safety Unit | *ATmega2560*

- Controlled battery, inverter, charging and auxiliary loads over 64 I2C channels (4x TCA9555) with pre-charge sequencing and contactor welding diagnostics.
- Implemented **sub-10 ms** emergency disconnect on HVIL breach and JSON diagnostics to the vehicle control unit.

ADDITIONAL PROJECTS

- **IoT Bus Ticketing & Fleet Tracking (EC200U, QuecPython):** UART QR scanning, mutual-TLS MQTT to Azure IoT Hub, GNSS tracking, offline-first cache for zero ticket loss.
- **TinyUSB Host (STM32H723, Cortex-M7):** MSC/HID host, MPU non-cacheable DMA memory (L1 cache fix). **Modbus RTU Master/Slave (STM32H533):** RS-485 DE, CRC-16.
- **Surgical Light Controller (STM32H503):** GPDMA thermistor sensing, 70°C cutoff. **DAC-less Audio (PY32F002B):** TIM1 PWM DAC, TIM14 44.1 kHz clock.
- **Also:** Dual LiDAR Geotagged Logger (ESP32), RFID Multi-Door Security (PY32), Stepper Controller, Line-Following Robot (STM32G0).

EDUCATION

B.E. Electronics & Communication | Government Engineering College, Gandhinagar *April 2024 | CGPA 8.16*